Agentic Systems
in Biomedicine

An introduction

RSE

Research Software Engineering (RSE)

rse.cuni.cz

RSE

- **Lowers the barrier of researchers to computing tools and infrastructures**
- Facilitates the access to HPC computing
- We consult and develop software solution for research
- Domain expertise (Bioinformatics, HPC, Digital Humanities, etc)

Typical use cases

- I need to **automate** processing of medical imaging data (e.g. volumetry, segmentation).
- I need to use a **specialized infrastructure** for working with sensitive data (SensitiveCloud) but I need assistance setting it up.
- I am preparing a **research proposal** with a computing component, but I need to consult my ideas or requirements.
- I need **help designing** an online platform for collection of research data from clinicians.
- I need to d**eploy my software** on different computing infrastructures.
- I **need to help with setting up multi-agentic workflow**

Agents and agentic systems

- **AI agent**: a system that uses a language model to pursue a goal
- It can **plan steps**, **choose tools**, **execute actions**, and **use feedback** from the environment
- Typical tools: code execution, file access, databases, web search, APIs, workflow managers, HPC/Slurm jobs
- **Agentic system**: an architecture where one or more agents interact with tools, data, users, and each other

- An agent can **act**. Chatbot generates text
- Automate complex workflows, **connect reasoning with computation**, and reduce manual orchestration
- Main risk: incorrect actions, unsafe tool use, data leakage, uncontrolled side effects (e.g. *“delete files after you finish”*),Safe deployment requires **bounded permissions**, logging, human approval for critical actions, reproducibility, and traceability

Example Dataset

1,000 synthetic patient records with 7 variables and a built-in confounder

- Variables include **sex, age, episodes, infarction, antihypertensives, naive CD4**
- Roughly balanced by sex, no missing values
- Built-in patterns: **naive CD4 decreases with age**, antihypertensive use increases with age, infarction more frequent in men
- Main question: **Do antihypertensives affect naive CD4 T-cell levels?**
- A naive comparison suggests an effect, but **age is the key confounder**

> You are a statistical assistant for biomedical research. You work in R. After a dataset is uploaded, you briefly summarize it and then wait for questions from the physician/researcher. Your responses will contain a single chunk of R code that can be copy-pasted, along with a brief commentary on the results.

Thought for 5s

Understood.

Once you upload a dataset, I will:

- briefly summarize its structure and key variables,
- then wait for your specific research/statistical question,
- and answer with exactly one copy-pasteable R code chunk plus a brief interpretation.

Upload the data when ready.

Data Upload + Summary

The LLM inspects the uploaded table, identifies variable types, and reports a clean summary

I'll inspect the uploaded table and give you a short data summary first.

Thought for 20s >

R

```r
df <- read.csv("data130425.csv", sep = "\t")

str(df)
summary(df)
colSums(is.na(df))
table(df$sex, useNA = "ifany")
table(df$episodes, useNA = "ifany")
table(df$infarction, useNA = "ifany")
table(df$antihypertensives, useNA = "ifany")
```

Dataset summary: 1000 rows and 7 variables. The variables appear to be `pac` (ID), `sex`, `episodes`, `infarction`, `antihypertensives`, `naive_cd4`, and `age`. The table looks clean at first glance, with no obvious missing values in the uploaded file. `sex` is c[illegible]orical; the others are numeric, with `episodes`, `infarction`, and `antihypertensives` likely binary indicators. Ask your first research question and I'll continue with one R chunk plus a short interpretation.

Research Question + Initial Analysis

User asks about antihypertensives and CD4 levels — the model runs crude + adjusted analyses

How does adminsuatration antihypertensives drugs influence the levels of naive cd4 t cells?

I’m checking the relevant variables and will give you a single R chunk for the comparison.

Thought for 25s ›

R

```r
# crude comparison
aggregate(naive_cd4 ~ antihypertensives, data = df, function(x) {
  c(n = length(x), mean = mean(x), sd = sd(x), median = median(x))
})

t.test(naive_cd4 ~ antihypertensives, data = df)

# adjusted analysis
fit <- lm(naive_cd4 ~ antihypertensives + age + sex, data = df)
summary(fit)
confint(fit)
```

Crude comparison suggests that patients receiving antihypertensives have lower naive CD4 levels on average than untreated patients. But after adjustment for age and sex, the coefficient for `antihypertensives` is small and not statistically significant, while `age` shows a clear negative association with naive CD4. In this dataset, the apparent unadjusted difference is therefore most likely explained mainly by age rather than by antihypertensive treatment itself.

Detailed Results + Interpretation

After adjustment for age and sex, the apparent treatment effect vanishes — age is the confounder

include detailed results (including p-values in your response)

I'm extracting the actual estimates and p-values so I can report them concretely, not just generically.

Thought for 28s ›

R

```r
# Descriptive statistics by treatment group
aggregate(naive_cd4 ~ antihypertensives, data = df, function(x) {
  c(n = length(x),
    mean = mean(x),
    sd = sd(x),
    median = median(x))
})

# Unadjusted comparison
t.test(naive_cd4 ~ antihypertensives, data = df)

# Adjusted model
fit <- lm(naive_cd4 ~ antihypertensives + age + sex, data = df)
summary(fit)
confint(fit)
```

Detailed results: untreated patients (`antihypertensives = 0`) had mean naive CD4 of about **69.5** (SD **13.4**, n = **566**), while treated patients (`antihypertensives = 1`) had mean **66.3** (SD **13.0**, n = **434**). The unadjusted difference was therefore about **3.2 units lower** in treated patients, and this was statistically significant by Welch t-test (**p = 0.00013**).

After adjustment for `age` and `sex` in a linear model, the effect of antihypertensive treatment was no longer statistically significant: coefficient for `antihypertensives` about **+0.51**, **p = 0.498**, with 95% CI approximately **-0.97 to 1.98**. By contrast, `age` was strongly associated with lower naive CD4: coefficient about **-0.38 per year**, **p < 0.001**. `sex` was not significant (**p = 0.654**).

Interpretation: in this dataset, antihypertensive treatment appears associated with lower naive CD4 only in the crude analysis; after accounting for age and sex, there is **no evidence of an independent treatment effect**. The apparent crude difference is mainly explained by confounding, especially by age.

Agent-like Behavior of standard LLM

There's nothing new under the sun

Capabilities

- Can analyze uploaded data and answer follow-up questions
- Can generate code, run reasoning steps, and revise conclusions
- Can detect simple statistical pitfalls, such as confounding
- Basically **it can do things**

Limitations

- Limited context over longer workflows
- Cannot reliably handle large datasets or direct hardware access
- Weak persistent memory and no true agent-to-agent coordination
- No true multi-agentic capabilities

SYSTEM PROMPT USED FOR THE DEMO

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Commercial Solutions Exist

An Example: Kosmos AI for Scientific R&D

(Edison Scientific www.edisonscientific.com)

- Installs complex tools, analyzes data, etc.
- Spawns multiple agents working on subproblems
- Integrates the results
- Tests multiple hypotheses at the same time

Limitations

- Expensive (difficult to trains oneself to prompt correctly)
- Limited data storage
- *If only could we run something like that on our own (university) HW…*

Agents can be dangerous

I mean this as an earnest warning, not just a disclaimer: be very cautious what you allow an agent to do!

Basically an agent with access to a tool like *Python interpreter* can do whatever the user can do!

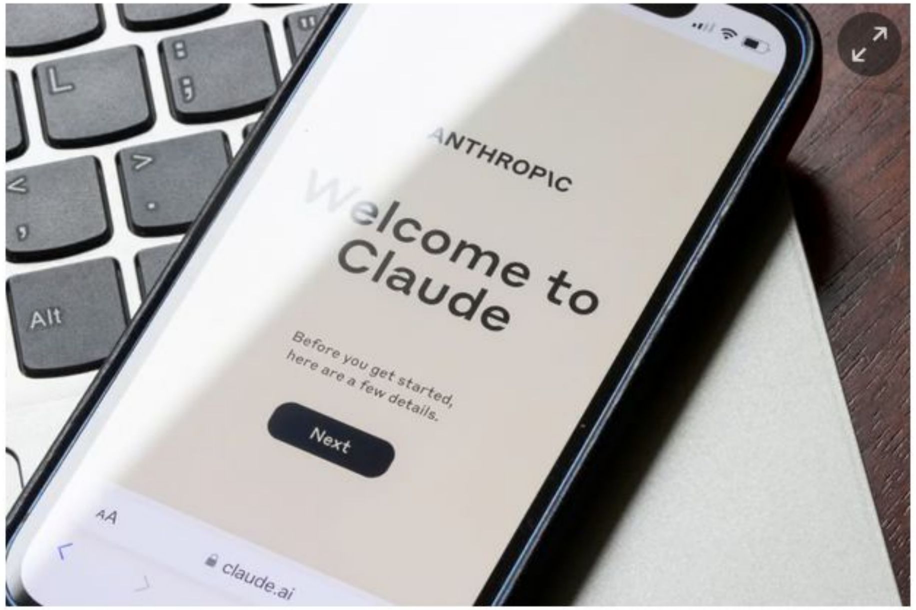
- Delete files (like…. all of them)
- Send information outside
- Delete databases
- Sending emails (receiving email, clicking, double-clicking,...)

(Nonexhaustive) Warnings

- Beware of third-party solutions (fraudulent MCP servers, frameworks, pipelines,...)
- Check the (user's) permission and access rights
- Experiment in a safe environment - e.*g. create a new user with limited access*
- Do not give access to the CLI or interpreters lightly
- Sandbox whenever possible

Claude-powered AI agent's confession after deleting a firm's entire database: 'I violated every principle I was given'

PocketOS was left scrambling after a rogue AI agent deleted swaths of code underpinning its business



The AI coding agent's destructive escapade left PocketOS' clients stranded. Photograph: Ted Hsu/Alamy

"It only took nine seconds for an AI coding agent gone rogue to delete a company's entire production database and its backups, according to its founder."

The Guardian (2026)

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